

CHE 201 Spring 2020 HW 3

Anna Christensen

TOTAL POINTS

65 / 100

QUESTION 1

1 Question 1 10 / 20

- 5 pts Wrong equation in convection
- 5 pts Wrong calculation in convection
- ✓ - 5 pts Wrong equation in radiation
- ✓ - 5 pts Wrong calculation in radiation
- 0 pts correct

QUESTION 2

2 Question 2 5 / 20

- 0 pts Correct
- 5 pts Wrong work of the gas
- ✓ - 5 pts Wrong heat transfer of the gas
- ✓ - 5 pts Wrong work of the piston
- ✓ - 5 pts Wrong potential energy of the piston

QUESTION 3

3 Question 3 15 / 20

- 0 pts Correct
- 20 pts Please solve the revised Problem 2.49 posted on the blackboard.
- ✓ - 5 pts $W_{\text{paddle}} = -3$ not 3 Btu. Paddle wheel transfers energy to the gas.
- 5 pts Wrong calculation

QUESTION 4

4 Question 4 20 / 20

- ✓ - 0 pts Correct
- 5 pts Wrong shaft work
- 5 pts Wrong displacing atmosphere work
- 5 pts Wrong heat transfer to the gas
- 2.5 pts Incomplete accounting of the heat transfer of energy
- 5 pts Wrong accounting of the heat transfer of energy

QUESTION 5

5 Question 5 15 / 20

- 0 pts Correct
- 1 pts Wrong sketch of the cycle
- 5 pts Wrong sketch of the cycle
- ✓ - 5 pts Wrong heat transfer for process 1-2
- 5 pts Wrong V_2
- 5 pts Wrong W_{cycle}

①



$$T_0 = 27^\circ\text{C}$$

$$h = 10 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$T_s = 47^\circ\text{C}$$

$$\epsilon = 0.93$$

$$\text{heat transfer coefficient} = 10 \frac{\text{W}}{\text{m}^2 \cdot \text{K}}$$

$$\dot{Q} = \epsilon \sigma A [T_0 - T_s]$$

$$T_0 = 27 + 273 = 300\text{K}$$

$$T_s = 47^\circ\text{C} + 273 = 320\text{K}$$

$$\frac{\dot{Q}}{A} = 0.93 \left(10 \frac{\text{W}}{\text{m}^2 \cdot \text{K}} \right) [300 - 320\text{K}]$$

$$\frac{\dot{Q}}{A} = -186 \frac{\text{W}}{\text{m}^2} \cdot \frac{1\text{KW}}{1000\text{W}} = \boxed{-0.186 \frac{\text{KW}}{\text{m}^2}}$$

②



$$a) V_1 = 0.1\text{m}^3$$

$$P = 2\text{bar}$$

$$P_2 = 0.12\text{m}^3$$

$$\Delta U = 0.25\text{KJ}$$

$$P = 2\text{bar} \cdot \frac{1 \times 10^5 \text{Pa}}{1\text{bar}} = 2 \times 10^5 \text{Pa}$$

$$W = \int P dV$$

$$\Delta U = Q + W$$

$$0.25\text{KJ} = Q + W$$

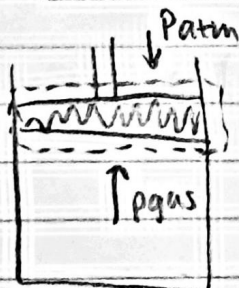
$$W = 2 \times 10^5 \text{Pa} (0.12\text{m}^3 - 0.1\text{m}^3)$$

$$W = 4 \times 10^3 \text{J} \cdot \frac{1\text{KJ}}{1000\text{J}} = \boxed{4\text{KJ} = W}$$

$$0.25 = Q + 4\text{KJ}$$

$$\boxed{Q = -3.75\text{KJ}}$$

b)



$$W = \int P dV$$

$$W = (\Sigma P)(dV) = (1\text{bar} + 2\text{bar})(0.12\text{m}^3 + 0.1\text{m}^3)$$

$$W = 0.06\text{bar} \cdot \text{m}^3 \times \frac{1 \times 10^5 \text{Pa}}{1\text{bar}} \times \frac{1\text{KPa}}{1000\text{Pa}}$$

$$P_{\text{atm}} = 1\text{bar}$$

$$P_{\text{gas}} = 2\text{bar}$$

$$\boxed{W = 6\text{KJ}}$$

$$\Delta U = -W$$

$$\boxed{\Delta U = -6\text{KJ}}$$

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- ✓ - 5 pts Wrong calculation in radiation
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$$\Delta U = Q + W$$

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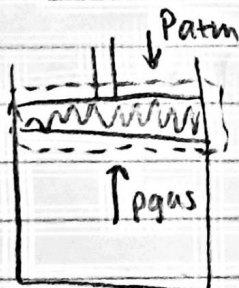
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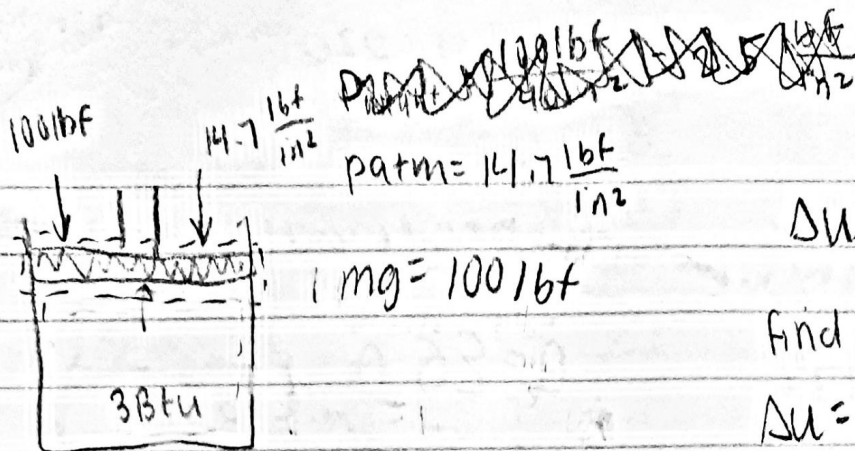
$$\Delta U = -W$$

$$\boxed{\Delta U = -6\text{KJ}}$$

2 Question 2 5 / 20

- 0 pts Correct
- 5 pts Wrong work of the gas
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- ✓ - 5 pts Wrong potential energy of the piston

③



$$\Delta U = 2.12 \text{ Btu}$$

Find ΔV

$$\Delta U = W_a + W_p + Q$$

$$2.5 \frac{\text{lbf}}{\text{in}^2} \times \frac{6894.76 \text{ Pa}}{1 \frac{\text{lbf}}{\text{in}^2}} = 17236.9$$

$$P_{\text{weight}} = 17236.9 \text{ Pa}$$

$$W_a = 1 \times 10^5 \text{ Pa} (V_2 - V_1)$$

$$W_p = 17236.9 (V_2 - V_1)$$

$$W = F \Delta h$$

$$\Delta h = \frac{W}{F}$$

$$\Delta PE = mg \Delta h$$

$$\Delta h = \frac{\Delta PE}{mg}$$

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$\Delta PE_{\text{piston}} + \Delta U_{\text{gas}} = -W$$

$$[F_{\text{piston}} + mg](z_2 - z_1) = -W - \Delta U$$

$$[P_{\text{atm}} \cdot A + mg](z_2 - z_1) = (-3 \text{ Btu} - 2.12 \text{ Btu}) \times \frac{9338.023}{1}$$

$$[14.17 \frac{\text{lbf}}{\text{in}^2} \cdot 40 \text{ in}^2 + 100 \text{ lbf}](z_2 - z_1) = -4.78 \times 10^4$$

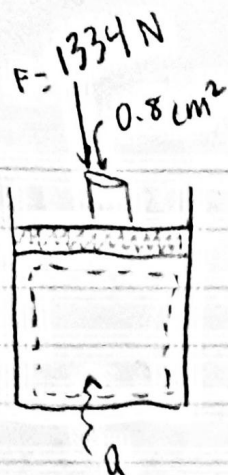
$$z_2 - z_1 = \frac{71.68 \text{ in}}{12} = \boxed{5.97 \text{ ft}}$$

3 Question 3 15 / 20

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- ✓ - 5 pts $W_{\text{paddle}} = -3$ not 3 Btu. Paddle wheel transfers energy to the gas.
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2.54

(4)



$$P_{atm} = 1 \text{ bar} = 1 \times 10^5 \text{ Pa}$$

$$m = 25 \text{ kg}$$

$$\Delta U = 0.1 \text{ kJ}$$

$$\Delta PE = 0.2 \text{ kJ}$$

$$d_{piston} = 10 \text{ cm} = 0.1 \text{ m}$$

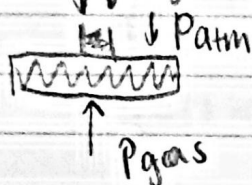
$$A_{piston} = \pi \left(\frac{0.1 \text{ m}}{2} \right)^2 = 7.85 \times 10^{-3} \text{ m}^2$$

$$A_{total} = A_{piston} - A_{shaft}$$

$$A_{total} = 7.85 \times 10^{-3} - 0.8 \text{ cm}^2 \left(\frac{1 \text{ m}}{1 \times 10^5 \text{ cm}^2} \right)$$

$$A_{total} = 7.77 \times 10^{-3} \text{ m}^2$$

$$1334 \text{ N} \downarrow \quad mg = 25(9.8) = 245 \text{ N} \downarrow$$



$$F_{\text{due to pressure}} = (P_{\text{gas}} - P_{\text{atm}}) \cdot A_{piston}$$

$$F_{\text{pressure}} = 1$$

$$a) \Delta PE = mg \Delta h$$

$$\Delta h = \frac{\Delta PE}{mg}$$

$$\Delta h = \frac{0.2 \text{ kJ}}{(25 \text{ kg})(9.8 \frac{\text{m}}{\text{s}^2})}$$

$$= 8.16 \times 10^{-4} \text{ m} = \Delta h$$

find work done by shaft

$$W = Fd = F \Delta h$$

$$W_{\text{shaft}} = (1334 \text{ N})(8.16 \times 10^{-4} \text{ m})$$

$$W_{\text{shaft}} = 1.1 \text{ kJ}$$

b) find work done by atmosphere

$$F_{\text{atm}} = 1 \times 10^5 \text{ Pa} (7.77 \times 10^{-3} \text{ m}^2) = 777 \text{ N}$$

$$W_{\text{atm}} = (777 \text{ N})(8.16 \times 10^{-4} \text{ m}) = 0.634 \text{ kJ} = W_{\text{atm}}$$

$$c) W_{\text{total}} = W_{\text{shaft}} + W_{\text{atm}} = 1.1 \text{ kJ} + 0.634 \text{ kJ} = 1.73 \text{ kJ}$$

$$\Delta U + \Delta KE + \Delta PE = Q - W$$

$$(\underbrace{\Delta U + \Delta KE}_{0} + \underbrace{\Delta PE}_{0})_{\text{gas}} + (\underbrace{\Delta U + \Delta KE}_{0} + \underbrace{\Delta PE}_{0})_{\text{shaft}} = Q - W$$

$$\Delta U_{\text{gas}} + \Delta PE_{\text{shaft}} = Q - W$$

$$0.11 \text{ kJ} + 0.2 \text{ kJ} = Q - 1.73 \text{ kJ}$$

$$\boxed{Q = 2.03 \text{ kJ}}$$

d)

amount	Energy input	energy stored		energy output	
	2.03 kJ	0.3 kJ		1.73 kJ	
		ΔPE_{shaft} 0.2	ΔU_{gas} 0.1	W_{shaft} 1.1	W_{atm} 0.634

4 Question 4 20 / 20

✓ - 0 pts Correct

- 5 pts Wrong shaft work
- 5 pts Wrong displacing atmosphere work
- 5 pts Wrong heat transfer to the gas
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(5)

process 1-2 $p = 1.4 \text{ bar}$ $W_{12} = 10.5 \text{ kJ}$
 $V_1 = 0.028 \text{ m}^3$

$$W = P(V_2 - V_1)$$

$$10500 \text{ J} = 1.4 \text{ bar} \left(\frac{1 \times 10^5 \text{ Pa}}{1 \text{ bar}} \right) (V_2 - 0.028 \text{ m}^3)$$

$$V_2 = 0.103 \text{ m}^3$$

process 2-3 $pV = \text{constant}$ $U_3 = U_2$
 $\Delta U = 0 \rightarrow \text{isothermal}$

$$W_{2-3} = P_2 V_2 \ln \left(\frac{V_3}{V_2} \right)$$

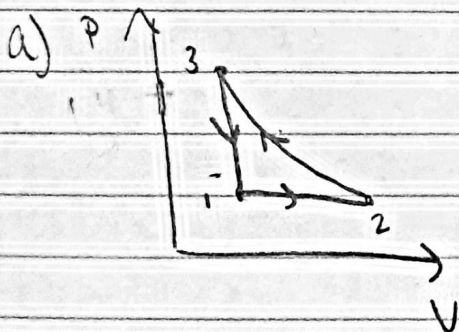
$$W_{2-3} = (1.4 \times 10^2 \text{ kPa}) (0.103 \text{ m}^3) \ln \left(\frac{0.028 \text{ m}^3}{0.103 \text{ m}^3} \right)$$

$$W_{2-3} = -18.78 \text{ kJ}$$

process 3-1 $U_1 - U_3 = -26.4 \text{ kJ}$
 constant V

$$W = P dV = W$$

$$W_{1-3} = 0$$



b) $W_{12} + W_{2-3} + W_{3-1} = W_{\text{total}}$

$$10.5 \text{ kJ} + -18.78 \text{ kJ} + 0 \text{ kJ} = -8.28 \text{ kJ}$$

c) $\Delta U = Q - W$

$$Q = \Delta U + W$$

$$\Delta U_{1-2} = ?$$

$$\Delta U_{2-3} = 0$$

$$\Delta U_{3-1} = -26.4 \text{ kJ}$$

$$Q = 24.6 \text{ kJ} - 18.78 \text{ kJ}$$

$$\Delta U_{1-2} + \Delta U_{2-3} + \Delta U_{3-1} = 0 \text{ for a cycle}$$

$$\Delta U_{1-2} + 0 - 26.4 = 0$$

$$Q = 5.82 \text{ kJ}$$

$$\Delta U_{1-2} = 26.4 \text{ kJ}$$

5 Question 5 15 / 20

- 0 pts Correct
- 1 pts Wrong sketch of the cycle
- 5 pts Wrong sketch of the cycle
- ✓ - 5 pts Wrong heat transfer for process 1-2
- 5 pts Wrong V_2
- 5 pts Wrong W_{cycle}